



Fig. 3: Ideally the circuit should have behaved as shown here

ditionally, I/O signal switching gives rise to switching noise on power rails. Having bypass capacitors helps bypass this high-frequency noise on the supply rails to system ground.

Selection of decoupling capacitors depends on the switching frequency components (frequency band) of interest. Designers should look at the capacitors' datasheets and the impedance versus frequency curves. Use only one value of the capacitor. Sometimes to extend the frequency range, designers are tempted to use two different values of capacitors that are purely based on the ideal model of a capacitor. But, practically, doing so creates problems at intermediate frequencies in the desired band.

Based on the ideal model of the capacitor, what is expected is depicted in Fig. 3. But what happens in reality is shown in Fig. 4. A practical model has an equivalent series inductor (ESL) and an equivalent series resistor (ESR). So when capacitors of two values are connected in parallel, a parallel resonant circuit (tank circuit) is formed, producing an unwanted response at intermediate frequencies. (Refer right side of Fig. 2, which has all the components required to form a resonant tank circuit.) Due to parallel resonance, the impedance for intermediate frequencies would increase and frequency components in this range may not get bypassed effectively.

**Ground bounce.** Ideally, the signal ground should be the same as the system ground. But in a practical system,

there are parasitic inductances involved (in this case, device pin and internal bond wires) and trace plus plane inductance. Under dynamic state, these cause the signal ground to jump above the system ground, until a steady state is reached. This ultimately means that the fall time of the signal

increases and other devices may see this signal falling down slower than expected.

The parasitic inductance is due to the lead package inductance and trace and plane inductance. A system designer could probably do nothing about the package lead inductance but certainly, he must do something about the latter.

When the output changes from 'HI' logic to 'LO' logic, the current flows through the bottom transistor, which is now 'on.' It then passes through the device and trace/plane inductance. This steep change in the flow of current generates potential drop across these two stray inductors. Ideally, the signal ground (REF) should have been the same as the system ground, without any potential difference between them. But due to these stray inductances REF remains at higher potential than the system ground. This is called ground bounce.

During this dynamic state, the output voltage remains higher, which may not be interpreted as 'LO' logic by the devices using this as an input, because their reference is the system ground. The ground bounce effect due to trace and plane inductances could be minimised by using heavy copper planes and bypass capacitors.

**The EMI (electromagnetic interference) concern.** Better signal integrity will produce a lower-EMI design. Though much gets revealed at the time of electrical design, care must be taken at the time of components selection.

Processors use external clock from sources like crystals and oscillators. If high-frequency oscillators are used, prefer spread-spectrum oscillators. EM emission generated at the oscillating frequency could cause the energy content to cross the allowable limit. The peak of this emitted energy could be brought down by using spread-spectrum oscillators, which vary the signal frequency within a small permissible range and distribute the total emitted energy content over a band of frequencies.

**Multiple power supplies—decoupling and sequencing.** Most electronic devices and processors use multiple supply voltages. The processor's core runs at lower voltage levels, which could be 2.5V, 1.8V, 1.1V or even less. Internal on-chip voltage regulators are used to vary this core voltage for dynamic power management. The I/O circuits usually employ 3.3V, while DDR-II I/Os use 1.8V.

Most processors have an on-chip phase-locked loop (PLL), which is used to derive higher-core and system clock frequencies from lower-input-clock source frequencies. The PLL again forms a part of dynamic power management and by programming the PLL at run-time to reduce core frequency, power consumption can be reduced—that's what is done when a system goes to standby mode from full-on mode.

It is important to understand if there is any power up/down sequencing requirement mentioned in the device datasheets. Additional external circuitry may be needed to meet such a sequencing requirement. In some cases, there may not be a functional requirement but there could be power implications in the sense that more power may be dissipated during the startup time when one supply comes up before the other. The default voltage level of I/Os may also get affected under certain supply sequencing conditions. These side-effects need to be well understood by the designers before computing the power budget and designing other system interfaces.